Self-Supervised SAR-Optical Data Fusion of Sentinel-1/-2 Images

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***Abstract*— The effective combination of the complementary information provided by huge amount of unlabeled multisensor data (e.g., synthetic aperture radar (SAR) and optical images) is a critical issue in remote sensing. Recently, contrastive learning methods have reached remarkable success in obtaining meaningful feature representations from multiview data. However, these methods only focus on image-level features, which may not satisfy the requirement for dense prediction tasks such as land-cover mapping. In this work, we propose a self-supervised framework for SAR-optical data fusion and land-cover mapping tasks. SAR and optical images are fused by using a multiview contrastive loss at image level and super-pixel level according to one of those possible strategies: in the early, intermediate, and late strategies. For the land-cover mapping task, we assign each pixel a land-cover class by the joint use of pretrained features and spectral information of the image itself. Experimental results show that the proposed approach not only achieves a comparable accuracy but also reduces the dimension of features with respect to the image-level contrastive learning method. Among three fusion strategies, the intermediate fusion strategy achieves the best performance. The combination of the pixel-level fusion approach and the self-training on spectral indices leads to further improvements in the land-cover mapping task with respect to the image-level fusion approach, especially with sparse pseudo labels. The code to reproduce our results will be found at https://github.com/yusin2it/SARoptical_fusion.**

***Index Terms*— Data fusion, land-cover mapping, pixel level, remote sensing, self-supervised learning, Sentinel-1/-2.**

I. Introduction

EVERY year, a large number of Earth observation satellites are operated to monitor human activities, Earth environment, and their mutual influences across our planet. Hundreds of terabytes of remote sensing (RS) data are accumulated per day from various systems, which cover most bands of the electromagnetic spectrum and include both active and passive sensors [1]. In this context, deep learning methods, especially supervised deep learning approaches, have been developed to process and analyze such massive amounts of multimodal RS data for specific applications, such as land-cover mapping, target recognition, and change detection. However, these applications are mostly limited to the use of a single type of image and require a large amount of labeled data for the training of the algorithm.

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The most common approach is based on deep learning techniques applied to single modality data, e.g., multispectral, hyperspectral, light detection and ranging (LiDAR), or synthetic aperture radar (SAR). The fusion of various RS data from different sensors has not received sufficient attention yet. However, it is well known that the complementary use of multimodal RS data offers more complete information on a scene and can result in better performance in many applications [2]. For example, multispectral/hyperspectral images acquire information that characterizes land-cover categories on the basis of their spectral signatures, while radar images provide dielectric properties and are not affected by cloud occlusions.

Inspired by the success of deep learning in computer vision (CV), some works [3]–[6] have investigated the fusion of multimodal RS data using deep learning methods. Their results have shown that deep learning techniques play a significant role in multimodal RS data fusion. However, the recent successes of deep learning techniques in multimodal RS data fusion mainly focused on supervised methods, which are often limited by the availability of annotated data. Labeled remote sensing data are often scarce. The limited access to such labeled data has driven the development of unsupervised methods, such as generative models (e.g., generative adversarial network (GAN) [7], convolutional autoencoder (CAE) [8], and variational autoencoder (VAE) [9]). Nevertheless, recent research [10] has shown that such generative models overly focus on pixels rather than on abstract feature representations.

For land-cover mapping tasks, existing supervised methods [6], [11], [12] often require prohibitive annotation effort for labeling data. Thus, semi-supervised techniques [13] have been developed and explored to train a classifier using very few labeled data. As a popular semi-supervised technique, the self-training approach trains the supervised classifier in various iterations with the help of a few labels and the large pool of unlabeled samples [14]. Besides, Chen *et al.* [15] have found that the use of self-supervised pretraining for fine-tuning in self-training can give a significant improvement in CV tasks.

Unlike semi-supervised learning, many unsupervised land-cover mapping works [16], [17] have used a set of spectral indices derived from SAR and optical images [e.g., normalized difference water index (NDWI), normalized difference vegetation index (NDVI), bare soil index (BI), and backscattering values (BS)] to select training samples from the image itself. These indices have the potential to highlight a set of land-cover classes in terms of their physical characteristics. We state that they can be used to select pseudo labels in a self-straining land-cover mapping paradigm.

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Recent research [18]–[22] in contrastive learning demonstrates how these methodologies can encourage the network to learn more interpretable and meaningful feature representations. This resulted in improvements in classification and segmentation tasks, where contrastive methods outperformed the generative counterparts. However, existing contrastive methods, such as multiview contrastive learning (MCL) [20], rarely consider the pixel-wise feature representations.

To address this limitation, we propose a new self-supervised approach to fuse the complementary information presented in SAR and optical images at the pixel level. The proposed approach can be implemented according to three pixel-wise fusion strategies: 1) early fusion (PixEF); 2) intermediate fusion (PixIF); and 3) late fusion (PixLF). We also investigate the self-trained land-cover mapping task by considering spectral information in SAR-optical images and the benefits of self-supervised pretrained features. The proposed SAR-optical fusion approach is compared with the instance-level contrastive method under the common linear protocol in the context of the land-cover mapping task. The evaluation of the proposed land-cover mapping approach was performed on a subset of DFC2020 dataset [23].

The main contributions of this work are as follows.

1) We first introduce and verify the effectiveness of multiview contrastive loss in SAR-optical data fusion. Then, we propose a self-supervised approach, which can obtain pixel-wise feature representations from SAR and optical image pairs without using any annotation. This is achieved by using U-Net [24] and the contrastive loss, by preserving local information at the superpixel level.
2) We compare different fusion strategies (i.e., early fusion, intermediate fusion, and late fusion) in the proposed approach. Concretely, late and intermediate fusion strategies learn feature representations by comparing SAR and optical images directly, whereas the early fusion strategy distills the complementary information from a concatenation of image pairs. In addition, the efficiency of SAR-optical fusion with respect to the use of a single modality in the land-cover mapping task is analyzed.
3) We conduct a self-trained land-cover mapping task by jointly using the spectral information of SAR-optical images and the pre-trained fusion features.

The rest of this article is organized as follows. Section II presents the related works of SAR-optical data fusion based on supervised and unsupervised methods. Section III introduces the proposed approach by describing the three considered self-supervised data fusion strategies and a self-trained land-cover mapping task. The descriptions of the dataset, network setup, and experimental settings are given in Section IV. Experimental results obtained on linear protocol and a land-cover mapping task are illustrated in Section V. Finally, Section VI concludes this article.

II. Related Works

Various feature fusion methods, including supervised learning and unsupervised learning techniques, have been investigated to improve the performance of combining complementary information from SAR and optical images. Early works already proved the effectiveness of combining SAR and optical data with the multilayer perception (MLP) classifier [25], [26]. Recent Sentinel-1 and Sentinel-2 images are combined to improve the land-use and land-cover (LULC) classification accuracy on monsoon regions using a random forest model in [27]. However, these fusion algorithms are a simple concatenation of SAR and optical images and have no capability to learn high-level features. With the development of deep convolution neural networks (CNNs), more and more works [6], [11], [12], [28], [29] have instead used CNNs to fuse SAR and optical images. Task-relevant features can be extracted from the input data automatically during the network training. Kussul *et al.* [28] first explored the deep CNNs in SAR-optical fusion for LULC classification and demonstrate their superiority with respect to traditional MLP classifiers. Feng *et al.* [29] proposed a multibranch CNN to improve the classification accuracy in coastal areas by fusing Sentinel-1 and Sentinel-2 images. A multitemporal W-Net was proposed to integrate Sentinel-1 and Sentinel-2 images in land-cover mapping [11]. An integration of fully convolutional networks (FCNs) and object-based methods was proposed for LULC mapping with the Sentinel SAR and optical images in [12]. Recently, Ienco *et al.* [6] proposed a deep learning architecture, namely TWINNS, to fuse Sentinel-1 and Sentinel-2 time series data in land-cover mapping. Adrian *et al.* [30] used the 3-D deep learning network to fuse multitemporal Sentinel-1 and Sentinel-2 data for mapping ten different crop types, as well as water, soil, and urban area. Besides the complete supervised fashion, Wang *et al.* [31] found a correlation between the image classification accuracy and the segmentation accuracy and demonstrated that a model trained on image classification performs well on a segmentation task in a weakly supervised way.

Supervised learning methods, despite their powerful ability to feature learning of SAR-optical images, are often limited by the amount of annotated data. An alternative method is to use unsupervised learning techniques, such as principal component analysis (PCA), probabilistic topic models, CAE, and canonical correlation analysis (CCA) [32]. These techniques can learn feature representations from unlabeled multimodal data. Amarsaikhan *et al.* [33] used PCA to enhance the features extracted from SAR-optical images and improve the urban land-cover maps. Fernandez-Beltran *et al.* [34] proposed a hierarchical multimodal probabilistic latent semantic analysis (HMpLSA) model to fuse SAR and multispectral imaging (MSI) data for unsupervised land-cover categorization tasks. Similarly, multiview learning methods also provide a solution to the unsupervised combination of complementary information from SAR-optical images. Geng *et al.* [35] proposed a deep bimodal autoencoder (BDAE) to fuse SAR and multispectral images for classification. Nielsen and Larsen [36] jointly analyzed Sentinel SAR and optical data for change detection using CCA. Based on CCA, Andrew *et al.* [37] proposed the deep CCA (DCCA), which learns separate representations for each modality from a shared latent subspace using CNNs.

Despite the success of these methods, their performance is limited in feature learning ability or overly focuses on

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the pixel values. The recent success of contrastive learning [19] shows the possibility of alleviating such a dependency and obtaining useful representations from unlabeled data. Tian *et al.* [20] proposed an extension of the contrastive loss to self-supervised learning at the multiview setting, where different views are related to the same context. An influential series of studies [19], [38], [39] of self-supervised learning reported significant improvements on downstream tasks, such as object detection and image classification with respect to their supervised learning counterparts. Instance discrimination, contrastive loss, and aggressive augmentation are the three key ingredients underlying these improvements [40]. In this context, Khosla *et al.* [41] further applied the contrastive approach to the fully supervised setting and further improved the performance of supervised results.

Even with the useful representation, it is not possible to perform a land-cover mapping task without labels. Many works [13], [16], [17] have been dedicated to decrease the use of labels using the spectral information in SAR-optical images themselves. Aswatha *et al.* [16] jointly used scattering and spectral properties from SAR and multispectral images to distinguish land-cover classes, which are used to generate training samples. Besides, they train a random forest classifier to predict land-cover classes based on the selected samples and learned features. Wu *et al.* [13] utilized the self-training algorithm to gradually assign highly confident pseudo labels to unlabeled hyperspectral images and employ spatial constraints to regulate the self-training process. Rather than training from scratch, Chen *et al.* [15] found that the self-training based on self-supervised pretraining can improve the performance in classification task.

III. Methodology

This section presents the proposed self-supervised approach to SAR-optical image fusion, which aims to learn pixel-wise representations from unlabeled SAR-optical image pairs. Like previous self-supervised works [19], [38], [39], the three key ingredients (i.e., instance discrimination, contrastive loss, and aggressive augmentation) are included in our work, where the shift transformation is used as a data augmentation approach in the contrastive paradigm. The proposed SAR-optical fusion approach consists of three fusion strategies: early fusion, intermediate fusion, and late fusion [see Fig. 1(a)–(c)]. Afterward, a land-cover mapping task based on the spectral information of the image itself and the pretrained features is presented.

A. Network Architecture

The proposed approach has two branches [Fig. 1(a)], where the input image of each branch has a relative shift. Each branch contains a ResUnet [42] block followed by a linear layer projector. After the projection, the same shift operation is performed on the output for feature alignment between two branches. We adopt a similar ResUnet architecture as in [42] and only use residual blocks in the encoder part. Like U-net, ResUnet consists of an encoder, a bridge, a decoder, and skip connections between the downsampling and upsampling paths. In this work, ResNet-18 is used as the encoder of the ResUnet block but without the fourth residual block. The decoder part has three blocks, where each block consists of a convolution layer (Conv) and a batch normalization (BN) layer, a rectified linear unit (ReLU) activation layer, and an upsampling operation. A 1×1 Conv, following with the last block, is used to reconstruct the learned representations. All the padding types in the ResUnet block were changed to the "same" padding.

TABLE I
Network Structure of the Proposed PixEF, PixLF, and PixIF

Name	PixEF	PixLF	PixIF
	Encoding	Encoding	Encoding 1&2
Conv1	[3×3, 64], stride 2	[3×3, 32], stride 2	[3×3, 32], stride 2
MaxPool	3×3, stride 2	3×3, stride 2	3×3, stride 2
ResBlk1 stride 2	$\begin{bmatrix} 3\times3, & 64 \\ 3\times3, & 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, & 32 \\ 3\times3, & 32 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, & 32 \\ 3\times3, & 32 \end{bmatrix} \times 2$
ResBlk2 stride 2	$\begin{bmatrix} 3\times3, & 128 \\ 3\times3, & 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, & 64 \\ 3\times3, & 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, & 64 \\ 3\times3, & 64 \end{bmatrix} \times 2$
ResBlk3 stride 2	$\begin{bmatrix} 3\times3, & 256 \\ 3\times3, & 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, & 128 \\ 3\times3, & 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times3, & 128 \\ 3\times3, & 128 \end{bmatrix} \times 2$
Bridge stride 1	[3×3, 256] upsampling 2	[3×3, 128] upsampling 2	[3×3,256] upsampling 2
	Decoding	Decoding	Decoding
Block4 stride 1	Cat. Block2 [3×3, 128] upsampling 2	Cat. Block2 [3×3, 64] upsampling 2	Cat. Block2 [3×3, 128] upsampling 2
Block5 stride 1	Cat. Block1 [3×3, 192] upsampling 2	Cat. Block1 [3×3, 96] upsampling 2	Cat. Block1 [3×3, 192] upsampling 2
Block6 stride 1	Cat. Conv1 [3×3, 256] upsampling 2 Conv 1×1	Cat. Conv1 [3×3, 128] upsampling 2 Conv 1×1	Cat. Conv1 [3×3, 256] upsampling 2 Conv 1×1

Feature alignment is achieved by using the same shift operation on both input images and output features of two branches. Specifically, given an input I_1 consisting of the SAR-optical image pair obtained from the same scene, we use a shift operation to make a random offset of I_1 along the x- and y-axis directions. In this way, we can define the random shift transformation as T and obtain an augmented view $I_2 = T(I_1)$. During the training, the augmented view and the original input are fed into two branches, respectively, to obtain pixel-wise representations v_1 and v_2. To align pixel-wise representations of two branches, the same transformation is applied to the output of the other branch $v_1 = T(v_1)$.

In particular, two branches of PixEF and PixIF share the same parameters, but the encoder of PixIF is split into two groups where each group has half channels of the counterpart of PixEF in each layer. We denote the network of PixEF as U_e and its encoder as E_e. Similarly, the network of PixIF as U_i and its two independent encoders as E_{i1} and E_{i2}. Unlike these two models, the PixLF has two independent branches with half channels of PixEF in each layer, where the input channels were adjusted to the input images. We denote the network of PixLF as U_l and its two independent encoders as E_{l1} and E_{l2}. The parameters and channel size of each unit are presented in Table I, where each convolution unit ([ ]) includes a convolutional layer, a BN layer, and a ReLU activation layer. Each residual block (ResBlk) in the encoding path has two

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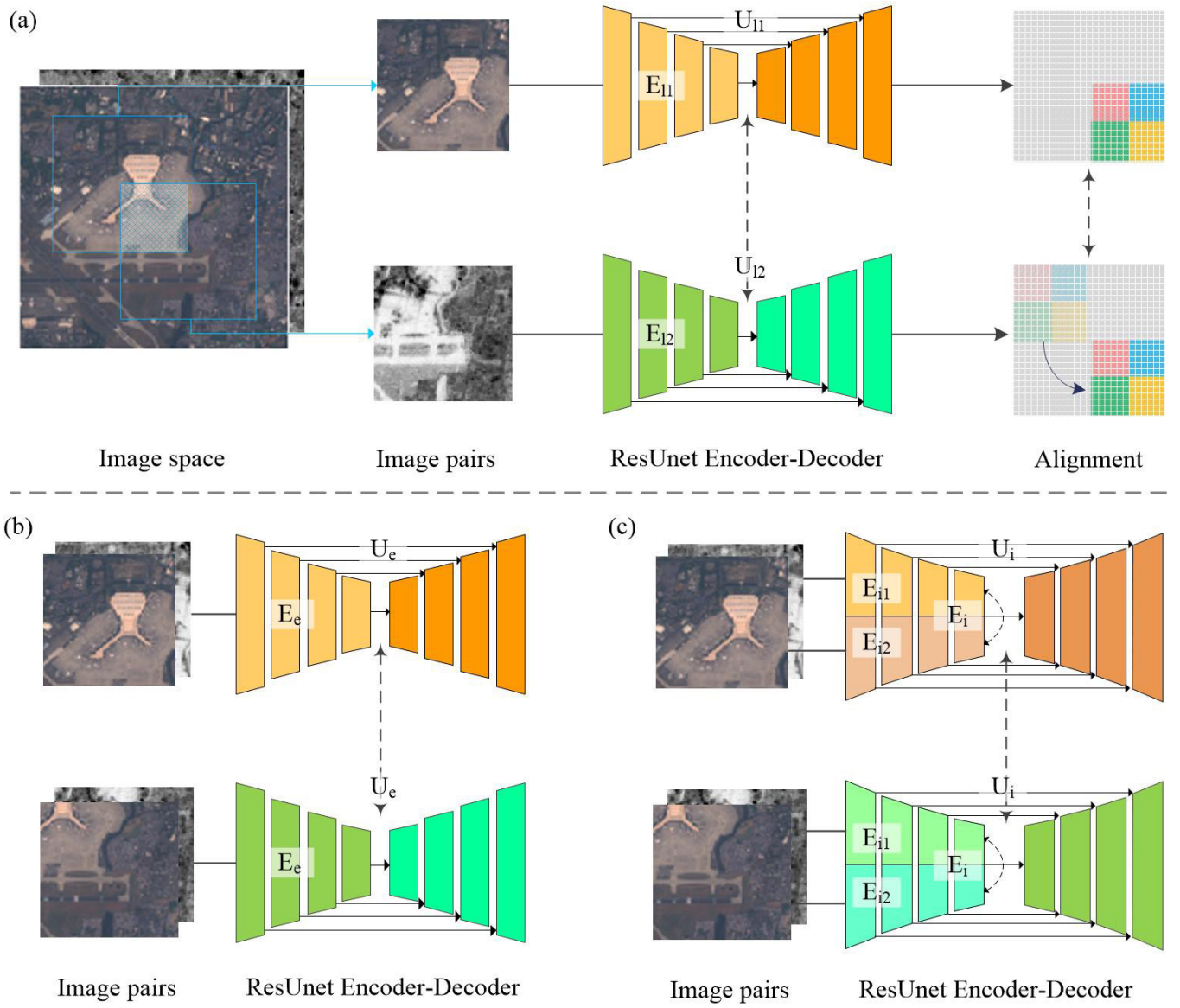


Fig. 1. Overview of the presented self-supervised SAR-optical fusion approach. The dashed arrow line represents a contrastive loss. (a) Illustration of pixel-wise representation learning framework for the late fusion strategy. The two inputs have an offset but keep an overlap. The approach follows the common contrastive learning architecture where both branches consist of a ResUnet block and a projection. Then, a shift transformation is included in the one branch for aligning representations between two branches. (b) ResUnet block follows the early fusion strategy. (c) ResUnet block follows the intermediate fusion strategy where the encoder contains two parts used for encoding SAR and optical images independently.

residual units. Each residual unit consists of two convolution units and an identity mapping.

B. Loss Function

The proposed approach consists of two types of contrastive loss based on images and superpixels individually. The main idea behind a contrastive loss is to find a feature representation that is invariant to augmentations. Given a dataset S that consists of a collection of image pairs $\{(s_1^i, s_2^i)\}_{i=1}^N$ across N different scenes, we consider each image pair (s_1^i, s_2^i) sampled from the joint distribution $p(s_1^i, s_2^i)$, which we call positives. Let s_2^j be taken from another scene $(j \neq i)$, and then, samples (s_1^i, s_2^j) sampled from the product of marginals $p(s_1^i)p(s_2^j)$, which we call negatives. The model $h(\cdot)$ is expected to know which pair is drawn from the joint distribution, while the other is not exactly, by computing their cosine similarity with a hyperparameter τ. In the multiview setting, the model $h(\cdot)$ is a neural network consisting of two branches with independent or same parameter f_{θ_1} and f_{θ_2}

$$h(s_1, s_2) = \exp\left(\frac{f_{\theta_1}(s_1) \cdot f_{\theta_2}(s_2)}{\|f_{\theta_1}(s_1)\| \cdot \|f_{\theta_2}(s_2)\|} \cdot \frac{1}{\tau}\right) \tag{1}$$

where s_1 and s_2 are the inputs in two branches of the network. The final loss function can be written as $L(f_{\theta_1}, f_{\theta_2}, S)$ given the dataset S

$$L\left(f_{\theta_1}, f_{\theta_2}, S\right) = -\mathop{\mathbb{E}}_{S}\left[\log \frac{h\left(s_1^i, s_2^i\right)}{\sum_{j=1}^N h\left(s_1^i, s_2^j\right)}\right] \tag{2}$$

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where (s_1^i, s_2^i) is a positive pair sample, $(s_1^i, s_2^j | j \neq i)$ is a negative pair sample, and $\{s_1^1, s_2^1, s_2^2, \ldots, s_2^N\}$ is a set that contains $N-1$ negative samples and one positive sample by anchoring at s_1^1. In the training process, the network is trained to increase the value of positive pairs and decrease the value of negative pairs. This results in a feature representation that is close to positive pairs, whereas it is not appropriate for negative pairs.

In the pixel-level contrastive loss, we sample and average features from two branches over superpixels that are located on the overlap between the two branches. This aims to keep the consistency of the normalized pixel-wise representations between two branches. Here, we construct a set of pixel-wise feature pairs $P_{i=1}^N$ where the positive feature pair (p_1^i, p_2^i) is sampled from the same location, while $p_2^j | j \neq i$ in negative pairs is taken from another location. Compared with the instance-level contrastive learning, this loss function can make the model get more detailed representations and thus more suitable for dense prediction downstream tasks. To overcome the noise when using the single pixel, we adopt the contrastive loss at the superpixel level. Together with the pixel-level contrastive loss, an instance-level contrastive loss is used to improve the performance. The instance-level loss helps to discriminate the similarity between the shifted views. Like pixel-level loss, we can construct a set $M_{i=1}^N$ of concatenated image pairs, where (m_1^i, m_2^i) is sampled from the same scene i, while $m_2^j | j \neq i$ is taken from another scene. Finally, we use the pixel-wise contrastive loss in conjunction with the instance-level contrastive loss, leading to the total loss of three fusion approaches

$$\begin{aligned} \mathcal{L}_e &= L(U_e, U_e, P) + L(E_e, E_e, M) \\ \mathcal{L}_l &= L(U_{l1}, U_{l2}, P) + L(E_{l1}, E_{l2}, M) \\ \mathcal{L}_i &= L(U_i, U_i, P) + L(E_{i1}, E_{i2}, M) + L(E_i, E_i, M) \end{aligned} \quad (3)$$

where $\mathcal{L}_l$, $\mathcal{L}_e$, and $\mathcal{L}_i$ are the loss functions of PixLF, PixEF, and PixIF, respectively.

C. Self-Training Land-Cover Mapping

To illustrate the use of the proposed SAR-optical fusion approach in land-cover mapping tasks that do not rely on manually annotated data, a self-training process is introduced and used in conjunction with the spectral information of images. We first calculated a set of spectral indices of each image and then selected reliable, representative, and diverse pseudo labels according to the characteristics of different land-cover classes. For different spectral indices, the larger index values the pixels have, the greater the possibility that pixels are associated with the related land-cover classes. In this article, NDVI, NDWI, BI, and BS were used to produce the index images for different classes. The generation of pseudo labels for different land-cover classes not only considers a certain spectral index but also the indices related to other classes, which can be utilized for synergistically labeling a pixel.

To avoid choosing a threshold value for each index directly, we used k-means [43] to over cluster Sentinel-1/-2 images and calculated the average spectral indices and average backscatter value for each class. Here, we segmented Sentinel-2 images into eight classes and Sentinel-1 images into four classes. Then, we define a set of generic rules (see Algorithm 1) for labeling six classes considering all indices and the combination of pseudo-classes defined by k-means on Sentinel-1 (Im_1) and -2 (Im_2) images. We denote labeled pixels of the forest, grassland, water, urban, bare land, and sparse vegetation as U^f, U^g, U^w, U^u, U^b, and U^s, respectively. For instance, the NDVI can indicate forest and grassland, and backscatter values in SAR images can further discriminate between these two classes. It is noted that just a few pixels with high confidence are labeled with a given class and the mean value of predefined classes is used as the threshold value.

Algorithm 1 Unsupervised Training Samples Collection

Input: Im_1, Im_2 images, NDVI, NDWI, BI and BS
Output: U^f, U^g, U^w, U^u, U^b, U^s

Overclustering Im_1, Im_2 into m and n classes.
Selecting the classes with maximun indicies values H_{ndvi}, H_{ndwi}, H_{bi} in Im_2 and H_{bs} in Im_1 as well as the class with minum value L_{bs} in Im_1. Also selecting the classes with medium NDVI and BI value: M_{ndvi} and M_{bi}. Then calculate their mean values V_{ndvi}, V_{ndwi}, V_{bi}, V_{bs}.
Iterating each pixel p in Im_1 and Im_2.
if p in H_{ndwi} and L_{bs}; NDWI$(p_2) > V_{ndwi}$ **then**
  $p \in U^w$
elif p in H_{ndvi} and H_{bs}; NDVI$(p_2) > V_{ndvi}$ **then**
  $p \in U^f$
elif p in H_{ndvi} and L_{bs}; NDVI$(p_2) > V_{ndvi}$ **then**
  $p \in U^g$
elif p in H_{bs}; BS$(p_1) > V_{bs}$ **then**
  $p \in U^u$
elif p in H_{bi} and L_{bs}; BI$(p_2) > V_{bi}$ **then**
  $p \in U^b$
elif p in M_{bi} and M_{ndvi}; NDVI$(p_2) < V_{ndvi}$ **then**
  $p \in U^s$
else
  $p \in$ None
end if
return U^f, U^g, U^w, U^u, U^b, U^s

The generated sparse pseudo labels are utilized to train a linear classifier layer based on the learned fusion features, and then, the classes of unlabeled pixels are predicted using the well-trained model. As the generated labels inevitably contain incorrect predictions, we fine-tune all parameters on the inference maps. The training framework in this task is shown in Fig. 2, where the pretrained PixIF model is used as the backbone followed by a linear classifier.

IV. Design of Experiments

In this section, we present the dataset for the training and validation of the proposed approach as well as the land-cover mapping task used for illustrating its application to a real task. Besides, the details of network setup and evaluation experiments are introduced.

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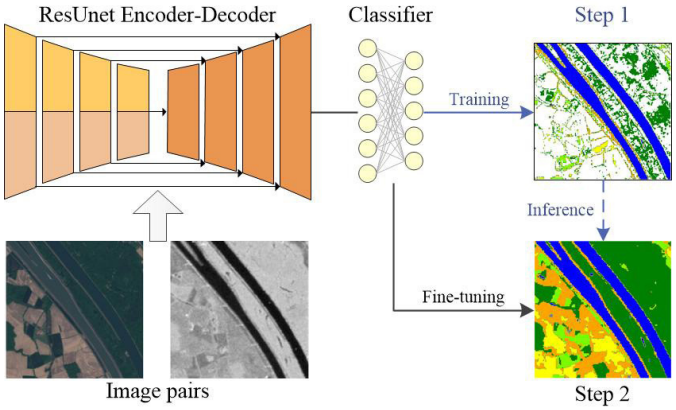


Fig. 2. Overview of the presented self-training land-cover mapping approach. Step 1 is the training on the sparse pseudo labels. Then, step 2 performs the training of whole network on the predicted labels.

A. Description of the Dataset

1) DFC2020: We developed our experiments on the DFC2020 dataset. The DFC2020, which has been issued by the IEEE-GRSS 2020 Data Fusion Contest [23], is used as the training set and the evaluation set for comparison between different methods. This dataset consists of a total of 6114 quadruple samples, which are SAR-optical image pairs, moderate resolution imaging spectrometer (MODIS)-derived labels, and more accurate semimanually derived high resolution (10 m) land-cover maps [44]. SAR images were acquired by Sentinel-1 and consist of dual-polarized (VV and VH) components, and the optical images were taken by multispectral Sentinel-2. Each SAR-optical image pair was obtained within the same season. Each pixel in the DFC2020 was assigned to a land-cover class manually, which has eight fine-grained classes (i.e., forest, shrubland, grassland, wetlands, croplands, urban/built-up, barren, and water). We also provide the image-level label for each image, which is derived by the majority class of the related pixel-level land-cover maps. Previous research [31] pointed out the effectiveness of training a CNN on image-level labels that can guide the weakly supervised model (WSL) to learn a powerful representation of images.

2) Training and Test Sets: A random split of the DFC2020 dataset into a training set (1000) and a test set (5114) was applied in this work. Here, the training set is used to tune the parameters of the linear protocol in the evaluating phase. The test set is used to validate the effectiveness of the features learned using different methods. To assess the effectiveness of these methods with limited labels, we randomly split the training set into five groups with 10, 50, 100, 200, and 1000 samples. Each small number group is a subsampled version of the corresponding full training set. Note that all self-supervised and unsupervised models were trained on unlabeled SAR-optical image pairs. In addition, we selected 2000 samples from the whole dataset for the self-training land-cover mapping task. We reclassify the cropland, wetland, and grassland as grassland and reclassify shrubland and barren as bared land. Other classes keep the same label as the original one. Therefore, this dataset only contains six classes (i.e., forest, grassland, urban, bare land, water, and sparse vegetation) according to the land surface properties.

B. Network Setup

The training process of the self-supervised approach includes three parts for PixEF, PixLF, and PixIF. For PixEF, SAR-optical image pairs were concatenated as one input. For PixIF and PixLF, SAR and optical images are in input to two branches independently. The Adam with a learning rate of $3e^{-4}$, a weight decay of $4e^{-4}$, and a momentum of 0.9 was adapted as our optimizer. We use a mini-batch size of 1000 with an input size of 16×16; models run for 700 epochs. We deploy a step scheduling learning rate policy in the training process. We use shift transformation to augment different inputs, where the vertical and horizontal ranges of the pixel shift are one-fifth of the input width. Apart from the shift transformation, we further apply a random flip transformation to improve the performance of the proposed approach.

For self-training in the land-cover mapping task, we train the model with a mini-batch size of 50, and models run for 100 epochs. The Adam optimizer is used with an initial learning rate of $1e^{-4}$ for the pretrained encoder layer and $3e^{-4}$ for the other parameters. When fine-tuning, the initial learning rate is set as $1e^{-4}$ for all parameters.

C. Experiment Settings

To evaluate the learned feature representation of different methods, we provide an evaluation with a linear classifier followed by the frozen features on the test set. In particular, the feature representation in the proposed approach has 256 channels, while that of the comparison methods (i.e., DCCA and MCL) is a concatenation of multilevel features with 512 channels. Note that we decided to use a linear classifier as our main evaluation metric for the quality of representations since it is simple and has a small number of extra parameters. This is an effective way to focus on the intrinsic discrimination capability of the classifier. The learning rate is set to 0.05 and the stochastic gradient descent (SGD) with a mini-batch size of eight was adopted as the optimizer for the linear protocol as well as the maximum number of epochs is 50.

In this experiment, we also assess the performance of the proposed approach applied to the Sentinel-1 image, the Sentinel-2 image, and both of them. This is done to validate whether the SAR-optical fusion can obtain more discriminative representations than single modality for the downstream land-cover mapping task. To this purpose, we trained the proposed approach only on single modality images in the proposed early fusion strategy. Then, the linear protocol on the frozen pretrained models was used to evaluate the effectiveness of learned representations with limited training labels.

In the land-cover mapping task, we adopt features obtained from the proposed self-supervised SAR-optical fusion approach and instance-level contrastive approach as pretraining. However, the instance-level self-supervised method does not have a decoder and cannot perform the same strategy. To provide a fair comparison, we also adopt the same decoder and the classifier of the proposed approach as a readout. The decoder is followed by encoders pretrained by instance-level self-supervised methods and used to reconstruct the concatenation features for downstream tasks. Furthermore, the proposed

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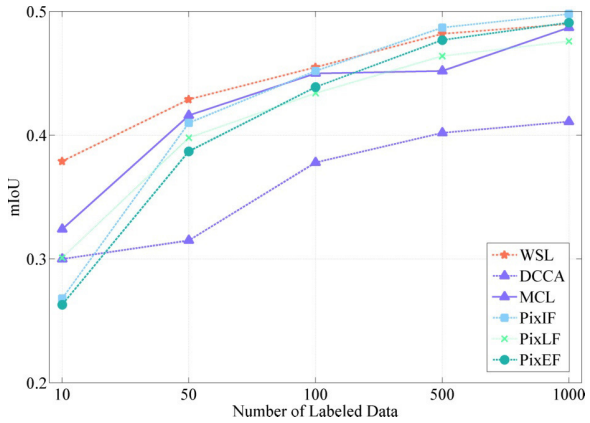


Fig. 3. mIoU metric achieved by different methods on test set versus the number of samples used for the training of the linear classifier on frozen encoders.

approach is fine-tuned with the predicted pseudo labels. To present a sparse label scenario, we just keep less than 10-pixel labels for each class of each image and discard the classes with less than 10-pixel labels.

V. Results

A. Evaluation With Linear Classifier

The performance of the proposed self-supervised approaches (PixEF, PixIF, and PixLF) was evaluated on the test set in comparison to the two instance-level self-supervised methods (DCCA and MCL) and the weakly supervised method (WSL). Also, in this case, we considered different amounts of labeled data for the training of the linear classifier (see Fig. 3). The average class accuracy (AA) and mean intersection over union (mIoU) are common metrics used to assess the performance in land-cover mapping and are used to evaluate the overall precision of all land-cover classes in our work.

Fig. 3 shows the linear protocol results on mIoU. As one can see that, the proposed PixIF outperforms all other methods, whereas the DCCA performs significantly worse against any other methods when the number of training samples increases. However, the WSL method, which is weakly supervised, outperforms all other methods with few labels. In general, the proposed PixEF and PixLF as well as the MCL have a similar performance. The gap between WSL and all contrastive approaches is reduced when the number of labeled samples increases. In particular, the proposed PixIF outperforms WSL in the case of 1000 training samples. Among contrastive approaches, the performance of PixIF and PixEF is slightly better than that of MCL. Moreover, it is worth noting that MCL obtained representations with 512 channels, while the proposed approaches are with only 256 channels. This means that the proposed approach significantly reduces the dimensionality of features while keeping or even improving the feature representation ability.

Table II presents a detailed comparison of the class-by-class accuracy obtained by the linear classifier on the test set for each approach when trained with 1000 labeled samples. According to the results, the proposed PixIF and PixEF as well as the weakly supervised WSL achieve an AA higher than 72% and an mIoU over 0.49 on the test set. They sharply outperform DCCA, which obtains an AA smaller than 65% and an mIoU smaller than 0.42. Among the proposed approaches, PixIF obtained the highest mIoU, whereas PixLF got the lowest value. In addition to the quantitative evaluations, we also provide a qualitative visual comparison of the land-cover maps predicted by different methods. Fig. 4 shows five examples of the results. Each example includes the land-cover maps predicted by different approaches as well as the ground truth in DFC2020. As one can observe that, the proposed self-supervised PixIF, PixEF, and the weakly supervised WSL show better results than the rest of the methods in all cases, and the proposed PixIF confirms to be more effective than the other two fusion strategies (PixEF and PixLF). For each land-cover class, similar conclusions to those derived in Table II can be given.

TABLE II
Class-Wise and Overall Accuracies Achieved on the Test Set by a Linear Classifier Used With the Different Methods Considering 1000 SAR-Optical Training Samples

Class	Average Accuracies (%)					
	WSL	DCCA	MCL	PixIF	PixLF	PixEF
Forest	90.2	92.7	90.9	**92.6**	92.3	91.3
Shrub-land	**70.4**	26.3	40.4	50.8	46.3	53.2
Grassland	57.6	63.0	68.7	73.0	64.5	**74.5**
Wetlands	**70.4**	56.9	64.9	62.7	57.2	59.9
Croplands	81.2	72.8	82.8	84.7	**85.8**	80.0
Urban	86.6	86.9	87.7	**87.8**	84.4	87.1
Barren	34.5	6.3	**46.3**	30.9	29.5	37.0
Water	99.2	99.0	99.3	99.3	**99.4**	99.2
AA	**73.8**	63.0	72.7	72.7	69.9	72.8
mIoU	0.490	0.411	0.487	**0.498**	0.476	0.490

In general, the results obtained in all comparisons confirm that the contrastive approach is superior to DCCA in this land-cover mapping task. The effectiveness of the proposed PixIF against other methods is due to its ability to use a three-level contrastive loss. It is interesting to note that despite the annotations are used in WSL, the proposed approach achieves comparable performance without any use of labels. This confirms the effectiveness of the self-supervised methods in feature representation learning.

We further investigate the performance of images from a single sensor in the land-cover mapping task, by training the proposed PixEF approach on Sentinel-1 (S1) and Sentinel-2 (S2) image as well as on the concatenation of Sentinel-1/-2 (S1S2) images. Table III shows a quantitative evaluation of the accuracy of each class by considering S1, S2, and S1S2 with the 1000 training samples and both linear and fine-tuning evaluations. One can see that the SAR-optical fusion outperforms the use of any single modality data in both linear protocol and fine-tuning evaluations. The performance of PixEF on S2 is very close to the performance on S1S2, while the performance on S1S2 has an increase of more than 10% of AA with respect to the performance on S1 in both types of evaluation. In addition, the results of Sentinel-2 images in higher classification accuracy on all classes than the use of only Sentinel-1 images for both evaluation methods. For individual classes, water achieved the highest accuracy

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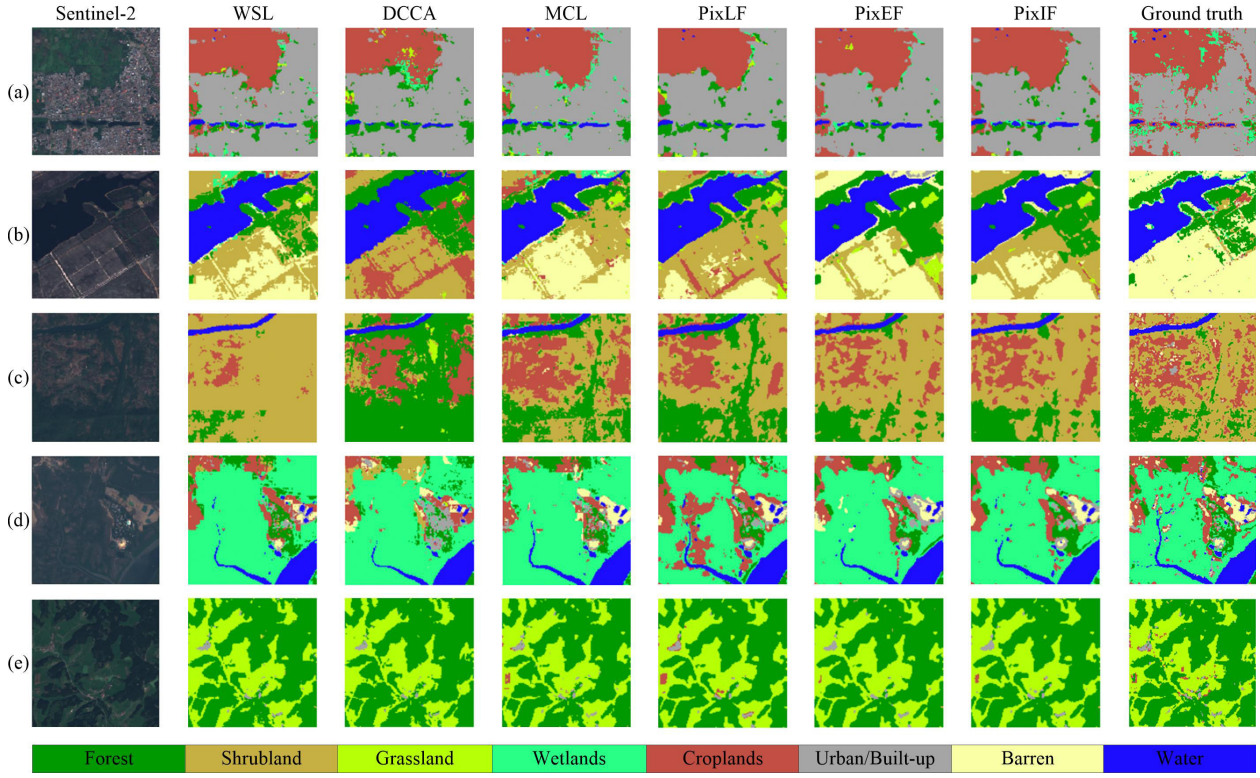


Fig. 4. Land-cover maps achieved on five different images by different considered methods with a linear classifier and the corresponding ground truths (see Table II for quantitative results).

TABLE III
CLASS-WISE AND OVERALL ACCURACIES ACHIEVED BY PIXEF ON SENTINEL-1 IMAGES ALONE (S1), SENTINEL-2 IMAGES ALONE (S2), AND SENTINEL-1/-2 IMAGE FUSION (S1S2) WITH THE LINEAR PROTOCOL AND THE FINE-TUNING EVALUATION

class	Linear Evaluation (%)			Fine-tuning Evaluation (%)		
	S1	S2	S1S2	S1	S2	S1S2
Forest	84.2	90.3	**91.3**	86.2	**92.2**	92.0
Shrubland	27.5	48.9	**53.2**	31.2	44.2	**62.3**
Grassland	61.1	67.2	**74.5**	61.9	68.0	**78.0**
Wetlands	35.0	58.8	**59.9**	51.5	62.3	**62.3**
Croplands	66.0	**81.4**	80.0	71.1	79.6	**85.7**
Urban	78.8	86.5	**87.1**	84.7	**89.7**	86.1
Barren	3.5	30.0	**37.0**	7.8	30.7	**38.6**
Water	98.8	99.2	**99.2**	99.2	**99.5**	99.4
AA	56.9	70.0	**72.8**	61.7	70.8	**75.6**
mIoU	0.362	0.470	**0.490**	0.395	0.474	**0.521**

in all conditions, whereas the barren has the lowest accuracy. The classification accuracy of water does not show an obvious improvement after fusion because there is already enough information in each single modality data. However, the SAR-optical fusion improved the performance of shrubland, grassland, and barren.

Apart from quantitative assessment, we also made a visual comparison of the results obtained with both evaluations on S1, S2, and S1S2 images. The performances of each modality and SAR-optical fusion keep consistent with both linear protocol and fine-tuning evaluations. As shown in Fig. 5, the SAR-optical fusion classifies various classes in a more accurate way, especially in barren, which obtains a significant improvement with respect to the use of single modality images. Besides, a trend can be figured out, that is, the methods with the input of S1S2 data achieve more smooth parsing results compared with the input of single modality data. Moreover, Fig. 5 also shows the advantages of each single modality and other improvements after SAR-optical fusion. Fig. 5(a) and (b) presents clouds on Sentinel-2 (but of course not in Sentinel-1) and the obvious influence on the corresponding classification maps. The presence of clouds leads to misclassifications in Sentinel-2, but not in Sentinel-1. Although the misclassification induced by clouds is still present in the classification maps of SAR-optical fusion, it is significantly mitigated compared to the results of Sentinel-2 alone. A similar phenomenon is also presented in Fig. 5(e), where the airport was distinguished as water in the Sentinel-1 result. This is the result of the similar backscatter between flat runway and water. Conversely, this was correctly distinguished as built-up in the Sentinel-2 result. After SAR-optical fusion, the classification errors in each modality were obviously reduced. Overall, the visual comparison is coherent with the quantitative results presented in Table III and confirms again the effectiveness of the presented self-supervised SAR-optical fusion approach.

B. Self-Trained Land-Cover Mapping Task

We further validate the proposed approach on a particular self-trained land-cover mapping task. Table IV presents a quantitative evaluation of the accuracy on each class obtained by the image-level feature fusion (MCL) and the pixel-level feature fusion (PixIF) as well as the fine-tuning of PixIF. As one can see, the pixel-level fusion approach achieves an

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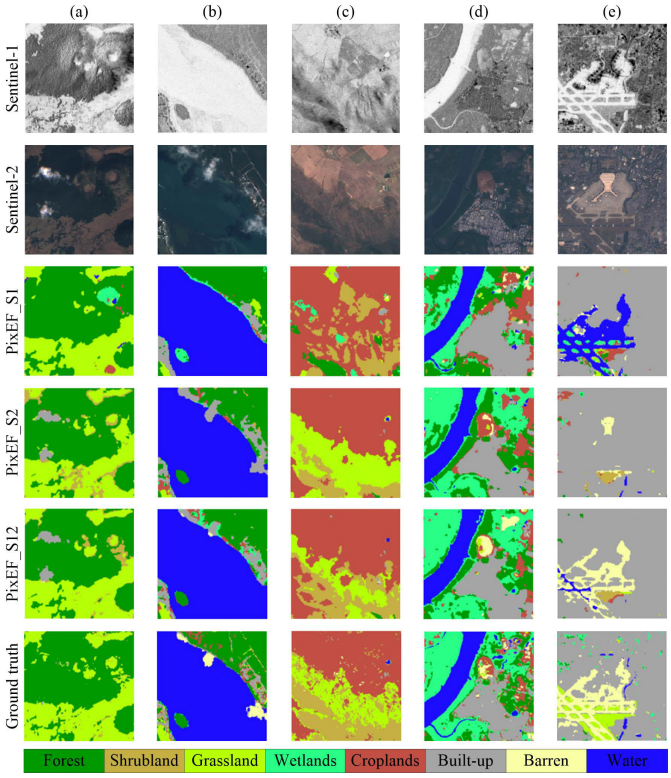


Fig. 5. Land-cover maps obtained by PixEF on Sentinel-1 images alone (S1), Sentinel-2 images alone (S2), and Sentinel-1/-2 image fusion with the linear classifier and fine-tuning evaluation for five different images and the corresponding ground truths (see Table III for quantitative results).

TABLE IV
CLASS-WISE AND OVERALL ACCURACIES ACHIEVED ON THE TEST SET BY A LINEAR CLASSIFIER USED WITH THE DIFFERENT METHODS CONSIDERING 2000 SAMPLES

Method	Forest	Sparse Veg.	Grass.	Built.	Barren	Water	AA	mIoU
MCL	87.3	36.9	53.3	68.8	42.5	99.0	64.7	0.408
PixIF	89.6	**39.3**	55.7	73.0	**47.4**	99.2	67.4	0.438
Fine.	**91.6**	38.2	**57.2**	**74.4**	47.1	**99.3**	**68.0**	**0.449**

AA of 67.4% sharply outperforming the image-level fusion approach, which obtains an AA of 64.7%. Similarly, compared with the image-level fusion, the pixel-level fusion outperforms it with a margin of 0.03 in mIoU. In particular, the fine-tuning obtains a higher AA and mIoU, with a significant improvement of 3.3 % and 0.04 with respect to the image-level fusion.

For individual classes, forest and water achieve the highest accuracy. In contrast, the accuracy of sparse vegetation and barren is below 50%. The classification of the sparsely vegetated ground is a big challenge for all methods. The possible reason is that the spectral characteristics of forest and water are very distinctive and very easy to be distinguished. On the contrary, barren and sparse vegetation present a high similarity of textural and spectral characteristics. Another possible reason reported in [45] is that there are also many misclassifications among shrubland, wetland, barren, and cropland contained in the origin ground truth. We can see from the results that the proposed approach can benefit the land-cover mapping task with sparse labels. Especially for the land-cover and crop-type mapping tasks, their labels are often extremely sparse, particularly in low- and middle-income regions [46].

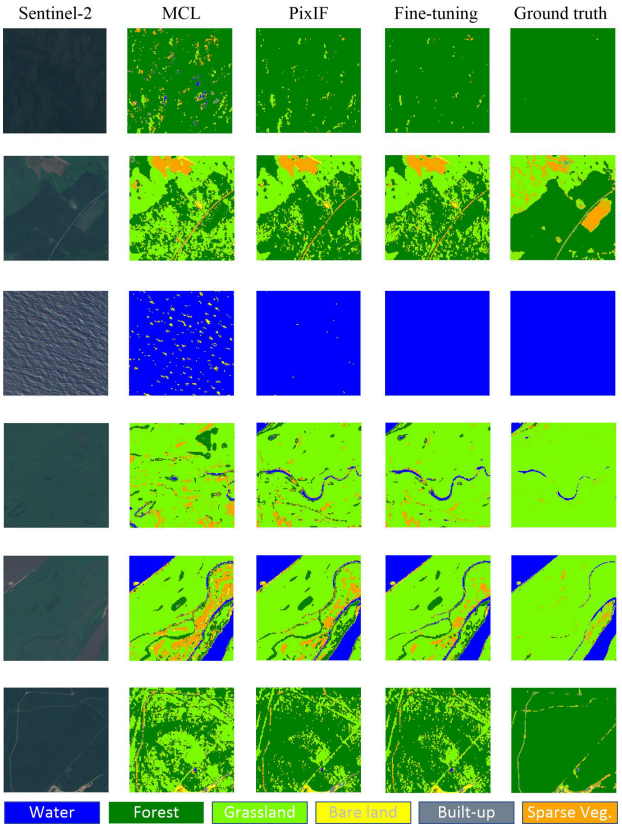


Fig. 6. Self-trained land-cover maps obtained by MCL with decoder, pixel-wise feature fusion (PixIF) and its fine-tuning results as well as the corresponding ground truths.

Apart from quantitative assessment, we also made a visual comparison of the land-cover maps predicted by different methods. As shown in Fig. 6, the pixel-level feature fusion classifies various classes in a more accurate way, especially in small areas. A trend can be figured out, that is, the proposed approach achieves smoother parsing results compared with the MCL ones. The result of fine-tuning also leads to a further performance improvement. Overall, the visual comparison is coherent with the quantitative results presented in Table IV and confirms again the effectiveness of the proposed approach on the land-cover mapping task.

C. Discussion

In this section, we discuss the effects of different components of PixIF that contribute to its performance. All results shown in Table V are trained and tested on the same setup of the linear protocol with 1000 training samples. The first row refers to the proposed approach using only shift operation and additional global loss. The second row refers to the proposed approach using geometric transformation (shift, rotate, resize, and sheer) instead of shift operation and additional global loss. The third row refers to the proposed approach using photometric transformation (gaussian blur and noise) and additional global loss. Also, the fourth row refers to the proposed approach only using shift operation.

As one can see, the combination of shift operation and the use of global contrastive loss achieves the highest accuracy.

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TABLE V
EFFECT OF THE USE OF GEOMETRIC, PHOTOMETRIC, SHIFT AUGMENTATION, AND GLOBAL LOSS IN THE PROPOSED APPROACH

Only Shift	Geometric	Photometric	Global loss	AA	mIoU
✓			✓	**72.7**	**0.498**
	✓		✓	68.4	0.460
		✓	✓	68.8	0.464
✓				70.6	0.479

In contrast, the lack of global loss makes the performance slightly decayed. This demonstrates the benefits of the use of instance-level contrastive loss. We also investigate a universal geometric transformation instead of shift operation in the network training. In this case, the performance drops about 0.04 on mIoU. The proposed approach can work with the geometric (affine) transformation but with little performance drops. Similarly, the photometric transformation leads to 0.03 drops on mIoU. In general, in this work, we found that the shift operation is the most useful and simple data augmentation approach for the proposed approach.

VI. CONCLUSION

In this article, we proposed a new self-supervised SAR-optical data fusion approach by jointly using the instance- and pixel-level contrastive loss and the shift data augmentation. The proposed approach explores three fusion strategies to distill the related representations from different modalities data. We additionally investigate the efficiency of SAR-optical fusion with respect to the single modality in the use of the proposed approach.

To evaluate the performance of the proposed approach, we compared it with two instance-level self-supervised methods (i.e., MCL and DCCA) and also with a weakly supervised method (WSL) considering the linear protocol evaluation with different numbers of training samples. The results show that the proposed PixIF achieves the best performance among all self-supervised methods and a comparable performance to that of a weakly supervised method. The effectiveness of the proposed PixIF can be explained by the use of different levels of contrastive loss for a dense prediction task. Comparisons between the performance of proposed PixEF on SAR-optical fusion and single modality data were also considered. The experiment confirmed again the benefit of SAR-optical fusion in the land-cover mapping task.

We also illustrate a self-training land-cover mapping approach based on the self-supervised pretraining and spectral information of images themselves. This approach is assessed quantitatively and qualitatively on the selected subset of DFC2020 and achieves an average accuracy of 68% considering six land-cover classes. Results show that the proposed PixIF together with the self-training approach can achieve a comparable result on land-cover mapping without manually annotated labels. The results confirm the benefit of the proposed approach in the land-cover mapping task with respect to the use of the instance-level self-supervised method. On the basis of the freely accessed Sentinel-1/-2 data, the proposed approach demonstrates a promising potential for automatic large-scale land-cover mapping. In addition, the proposed approach can also be used to fuse other raster data. However, it has also some limitations in the pseudo labels generation. The considered training strategy just gives six classes and focuses on backscatter values of Sentinel-1 images while ignoring the polarization information. Accordingly, as future development, we plan to explore the possibility of including more specific classes in the presented approach for the land-cover mapping task.

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